As per the Module-I Contents, the sample Q/A's are divided ~~into~~ as per the following ~~six~~ topics:

1. Basics (with AC quantities & 1-ϕ AC Ckt.) → Q1 & Q2
2. Phasor Notations / Phasor Diagrams → Q3, Q4, Q5
3. Series A.C. Ckt. → Q6
4. Parallel. A.C. Ckt. → Q7

   (3 & 4) + & Network Analysis in 1-ϕ AC Ckts → Q8, Q9, Q10
5. Series Resonance → Q11, Q13
6. Parallel Resonance. → Q12, Q14
7. Q - Factor. → Q13, Q14.

The Q/A's with these contents, are systematically presented below.

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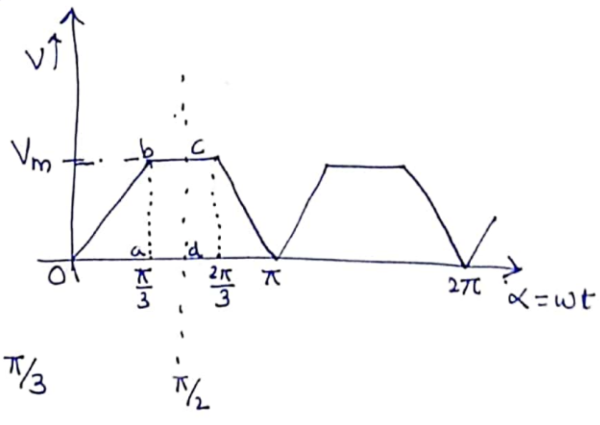
1) Basics :

Q1. Find the Avg. Value of the Periodic function of the waveform below (Fig. Aside)



A. From the curve, the waveform can be written as,

$$v(t) = \begin{cases} \dfrac{V_m}{\pi/3}(\omega t) = \dfrac{V_m}{\pi/3}\alpha & \text{when } 0 \le \alpha \le \pi/3 \\ V_m & \text{when } \dfrac{\pi}{3} \le \alpha \le \dfrac{\pi}{2}. \end{cases}$$

The given periodic function has time period 'T' $= \frac{\pi}{\omega}$ sec. It repeats at intervals of 'π' $= \pi$ radians.

Avg. value of the periodic $f^n = \frac{1}{T}\int_0^T f(t)\,dt$

(or)

$$F_{av.} = \frac{1}{T}\int_0^T f(\omega t)\,d(\omega t)$$ when 'T' is expressed in radians.

$\therefore$

$$V_{av.} = \frac{1}{\pi}\int_0^{\pi} v(\omega t)\,d(\omega t)$$

$$= \frac{1}{\pi}\left[\underbrace{\int_0^{\pi/3} \frac{V_m}{(\pi/3)}.\alpha\,d\alpha + \int_{\pi/3}^{\pi/2} V_m\,d\alpha}_{} + \underbrace{\int_{\pi/2}^{2\pi/3} V_m\,d\alpha + \int_{2\pi/3}^{\pi} \frac{V_m}{(\pi/3)}(\pi-\alpha)\,d\alpha}_{}\right]$$

wave symmetry

$$= \frac{2}{\pi}\left[\int_0^{\pi/3} \frac{V_m}{(\pi/3)}.\alpha\,d\alpha + \int_{\pi/3}^{\pi/2} V_m\,d\alpha\right]$$

$$= \frac{2}{\cancel{3}\pi}\left[\frac{V_m}{(\pi/3)}\left[\frac{\alpha^2}{2}\right]_0^{\pi/3} + V_m\left[\alpha\right]_{\pi/3}^{\pi/2}\right]$$

$$= \frac{2V_m}{\pi}\left[\frac{\pi}{6}+\frac{\pi}{6}\right] = \underline{\underline{\frac{2}{3}V_m}}$$

→ Result can be validated by finding area under curve from '0' to '$\pi/2$' & dividing by '$\pi/2$'.

$$Area\big|_{v(t)\text{ from '0' to '}\pi/2\text{'}} = Area\big|_{Oab} + Area\big|_{abcd} = \frac{1}{2}V_m\left(\frac{\pi}{3}\right)+V_m\left(\frac{\pi}{6}\right) = \frac{\pi}{3}V_m$$

$$\therefore\ V_{av} = \frac{area}{(\pi/2)} = \frac{(\pi/3)V_m}{(\pi/2)} = \underline{\underline{\frac{2}{3}V_m}}$$

Q2. A voltage $v(t) = 170 \sin(314t + 10^\circ)$ is applied to a circuit. It causes a steady-state current, $i(t) = 14.14 \sin(314t - 20^\circ)$ to flow in ckt. Find the Power Factor and Avg. Power delivered to the circuit.

A. Comparison of expressions $v(t)$ & $i(t)$ reveals that,

the phase-difference b/n v & i $= 30^\circ$

$\therefore$ Power Factor $= \cos\theta = \cos 30^\circ = \underline{\underline{0.866}}$

Also, $V_{rms} = \dfrac{V_{peak}}{\sqrt{2}} = \dfrac{170}{\sqrt{2}} = 120V$

$I_{rms} = \dfrac{I_{peak}}{\sqrt{2}} = \dfrac{14.14}{\sqrt{2}} = 10A.$

$\therefore$ Avg. Power / Active Power $= P_{av} = VI\cos\theta = 120 \times 10 \times 0.866$
$= \underline{\underline{1040W}}$

2) Phasor Notations/Diags:

Q3. The two sinusoidal currents are given as,

$i_1 = 10\sqrt{2}\sin\omega t \qquad i_2 = 20\sqrt{2}\sin(\omega t + 60^\circ)$

Find expression of sum of these currents using Phasor Addition.

A. 'i_1' & 'i_2' have RMS values of 10 A & 20 A

(Ref. Phasor)

i.e. $i_{1\,rms} = 10A \Rightarrow i_{1\,peak}$ / magnitude $= |I_1| = 10\sqrt{2}$; Angle $= 0^\circ$

$i_{2\,rms} = 20A \Rightarrow i_{2\,peak}$ / magnitude $= |I_2| = 20\sqrt{2}$; Angle $= 60^\circ$

Therefore, the phasor quantity of currents are,

$$\bar{I}_1 = 10\sqrt{2}\,(\cos 0^\circ + j\sin 0^\circ)$$

$$= \underline{\underline{10\sqrt{2}}}$$

$$\bar{I}_2 = 20\sqrt{2}\,(\cos 60^\circ + j\sin 60^\circ)$$

$$= 20\sqrt{2}\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = 10\sqrt{2} + j10\sqrt{6}$$

Thus, $\bar{I}_{net} = \bar{I}_1 + \bar{I}_2 = 10\sqrt{2} + (10\sqrt{2} + j10\sqrt{6})$

$$= 20\sqrt{2} + j10\sqrt{6}$$

$\Rightarrow$ $|I_{net}|$ (magnitude) $= \sqrt{(20\sqrt{2})^2 + (10\sqrt{6})^2} = \sqrt{800+600}$

$$= \underline{\underline{37.4}}$$

Angle of $\bar{I}_{net}$ i.e $\theta_{net} = \tan^{-1}\left(\frac{10\sqrt{6}}{20\sqrt{2}}\right) = \tan^{-1}\frac{\sqrt{3}}{2}$

$$= \underline{\underline{41^\circ}}$$

$\therefore$ $\underline{\bar{I}_{net} = 37.4\, e^{j41^\circ}}$

$$\Rightarrow \boxed{i_{net} = 37.4\sin(\omega t + 41^\circ)}$$

Note: In interest of simplicity, it is customary to replace $e^{j\theta}$ by $\angle\theta$; i.e $\bar{I}_{net} = 37.4e^{j41^\circ} = 37.4\angle 41^\circ$

Q4. A sinusoidal having effective/RMS value of $10\angle 0^\circ$ A is added to sinusoid current $20\angle 60^\circ$ A. Find resultant current.

A. $\bar{I}_{net} = \bar{I}_1 + \bar{I}_2 = 10\angle 0^\circ + 20\angle 60^\circ$

$$= 10 + (10 + j10\sqrt{3}) = 20 + j10\sqrt{3}$$
$$= 26.4\angle 41^\circ \text{ A}$$

So, for complete time expression,

$$i_{net} = \sqrt{2}\,.\,26.4 \sin(\omega t + 41^\circ)$$
$$= 37.4 \sin(\omega t + 41^\circ)$$

→ This is exactly equal to the i_{net} obtained in answer to Q3.

Note: In view of this, it is customary to use a phasor diagram in which the phasors are expressed as effective/RMS values, rather than using their maximum values.

Thus, phasor $10\angle 0^\circ \Rightarrow$ Magnitude of $10\sqrt{2}$, with RMS value 10A.
& " $\underset{\uparrow \text{RMS}}{20\angle 60^\circ} \Rightarrow$ " " $20\sqrt{2}$, " " 20A.

(This convention shall be followed in this course)

Q5. Evaluate the expression below using phasor method:

$$e(t) = 100\sqrt{2}\cos(314t - 30^\circ) \pm 200\sqrt{2}\sin(314t - 60^\circ)$$

For both '+' & '−' cases.

A. $e(t) = 100\sqrt{2}\cos(314t - 30^\circ) \pm 200\sqrt{2}\sin(314t - 60^\circ)$

Case
(I) $e(t) = 100\sqrt{2}\cos(314t - 30^\circ) + 200\sqrt{2}\sin(314t - 60^\circ)$
(II) $e(t) = 100\sqrt{2}\cos(314t - 30^\circ) - 200\sqrt{2}\sin(314t - 60^\circ)$

Case-I :

$$e(t) = 100\sqrt{2}\cos(314t - 30^\circ) + 200\sqrt{2}\sin(314t - 60^\circ)$$

$$= 100\sqrt{2}\cos(314t - 30^\circ) + \underbrace{200\sqrt{2}\cos\left(90^\circ - (314t - 60^\circ)\right)}$$

$$= 200\sqrt{2}\left(-\cos(314t + 30^\circ)\right)$$

$$\text{wt}\quad e(t) \equiv E\angle\theta = \bar{E}$$

$$\Rightarrow \quad \bar{E} = 100\angle{-30^\circ} - 200\angle{+30^\circ}$$

$$= 100\left(\frac{\sqrt{3}}{2} - j\cdot\frac{1}{2}\right) - 200\left(\frac{\sqrt{3}}{2} + j\cdot\frac{1}{2}\right)$$

$$= -100\cdot\frac{\sqrt{3}}{2} - 300\cdot\frac{1}{2}\cdot j$$

$$= -86.6 - 150j = 173.2\angle{-120^\circ}$$

$$\Rightarrow \boxed{e_1(t) = 173.2\sqrt{2}\cos(\omega t - 120^\circ)}$$

Case-II :

$$e(t) = 100\sqrt{2}\cos(314t - 30^\circ) - 200\sqrt{2}\sin(314t - 60^\circ)$$

$$= 100\sqrt{2}\cos(314t - 30^\circ) + 200\sqrt{2}\cos(314t + 30^\circ)$$

$$\therefore \quad \bar{E} = 100\angle{-30^\circ} + 200\angle{+30^\circ}$$

$$= 100\left(\frac{\sqrt{3}}{2} - j\cdot\frac{1}{2}\right) + 200\left(\frac{\sqrt{3}}{2} + j\cdot\frac{1}{2}\right)$$

$$= 300\,\frac{\sqrt{3}}{2} + 100\cdot\frac{1}{2}\cdot j$$

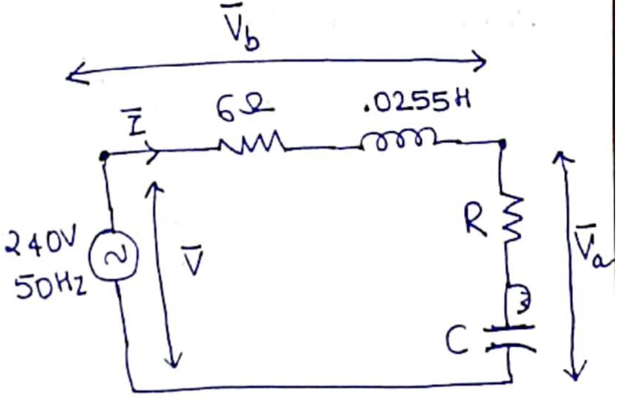
$$= 259.8 + 50j = 264.5\angle{10.53^\circ}$$

$$\Rightarrow \boxed{e_{II}(t) = 264.5\sqrt{2}\cos(\omega t + 10.53^\circ)}$$

*

3) Series AC Ckt:

Q6. In the circuit shown aside, Given that $\bar{V}_b$ & $\bar{V}_a$ are in quadrature, & $|V_b| = 3|V_a|$, Find 'R' & 'C'. Also depict the phasor relation between V, V_b, V_a & $\bar{I}$, with Phasor Diagram.



A. 240V, 50Hz Supply. $\Rightarrow |V|_{rms} = 240$

$L = .0255\ H$

$\Rightarrow X_L = \omega L$

$= 2 \times 3.14 \times 50 \times L$

$= 8\ \Omega$

It is said that phasors $\bar{V}_b$ & $\bar{V}_a$ are in Quadrature.

$$\Rightarrow |\bar{V}_a|^2_{rms} + |\bar{V}_b|^2_{rms} = |\bar{V}|^2_{rms} \quad \text{(from Fig.)}$$

& $|V_b| = 3|V_a|$

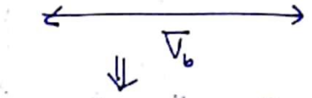
$$\Rightarrow |V_a|^2_{rms} + 9|V_a|^2_{rms} = |\bar{V}|^2_{rms}$$

$$\Rightarrow 10|V_a|^2 = 240^2 \Rightarrow |V_a|_{rms} = 75.9\ V.$$

$|V_b|_{rms} = 3|V_a|_{rms}$

$= 227.7V$

$\Downarrow$



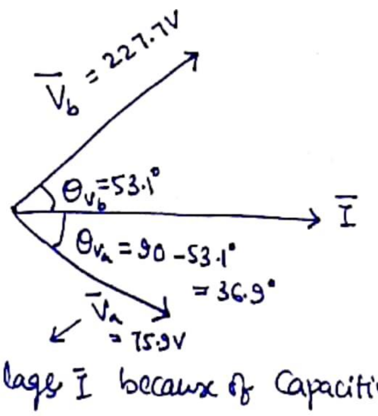
$\Downarrow$

$\bar{V}_b$ leads $\bar{I}$ ($\because$ Inductive element)

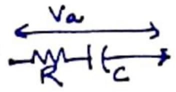
with angle, $\theta_{V_b} = \tan^{-1}\left(\frac{8}{6}\right)$

$= 53.1°$

$\therefore$ Phasor Diagram taking $\bar{I}$ as reference phasor:

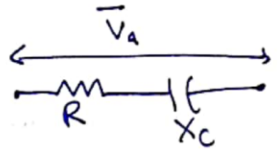


$\bar{V}_a$ lags $\bar{I}$ because of Capacitive element $\rightarrow$



$\therefore$ From the phasor above,

$$\theta_{V_a} = 36.9^\circ = \tan^{-1}\left(\frac{X_C}{R}\right)$$



$$\Rightarrow \frac{X_C}{R} = \tan 36.9^\circ$$
$$= 0.75 \quad \Rightarrow \quad \underline{X_C = 0.75R} \quad \text{—} (1)$$

Also, $|\bar{V}_b| = 227.7V \quad |\bar{I}| = \frac{|\bar{V}_b|}{|Z_L|} = \frac{227.7}{\sqrt{6^2+8^2}}$
$$= \underline{22.8}\ A$$

&

$$|\bar{I}| \times \cancel{\bar{V}_a}\ |Z_c| = |\bar{V}_a| \quad (\because \text{Series circuit})$$

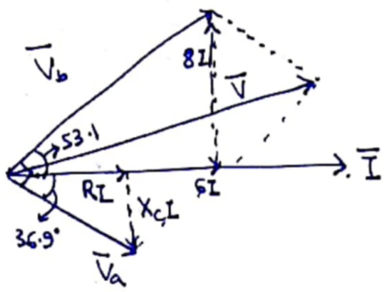
$$\Rightarrow 22.8 \cdot \sqrt{R^2 + X_C^2} = 75.9$$

$$\Rightarrow R^2 + X_C^2 = \left(\frac{75.9}{22.8}\right)^2 = 11.09 \quad \text{—} (2)$$

Solving (1) & (2), $\boxed{R = 2.664\,\Omega}$
$$X_C = 1.998\,\Omega$$

$$\& \ X_C = \frac{1}{\omega C} = \frac{1}{2 \times 3.14 \times 50 \times C} \Rightarrow \boxed{C = 1.594\ mF}$$

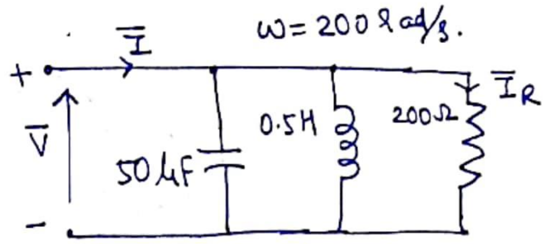
So,



(overall phasor Diagram)

4) Parallel AC Ckt :

Q7. In the parallel circuit Fig. aside,



(a) Find $\bar{I}$, if $\bar{I}_R = 0.02\angle 30^\circ$ A

(b) Find $\bar{I}_R$, if $\bar{I} = 2\angle -40^\circ$ A;
also find the applied voltage($\bar{V}$).

– Also, Identify, with associated phasor diagram, phasors $\bar{I}_R, \bar{I}_L, \bar{I}_C$ & $\bar{V}$ in (a).

A.

(a) Given $\bar{I}_R = .02\angle 30^\circ$ A.

$\Rightarrow$ Applied voltage, $\bar{V} = 200\Omega \times .02\angle 30^\circ$
$= 4\angle 30^\circ$ V.

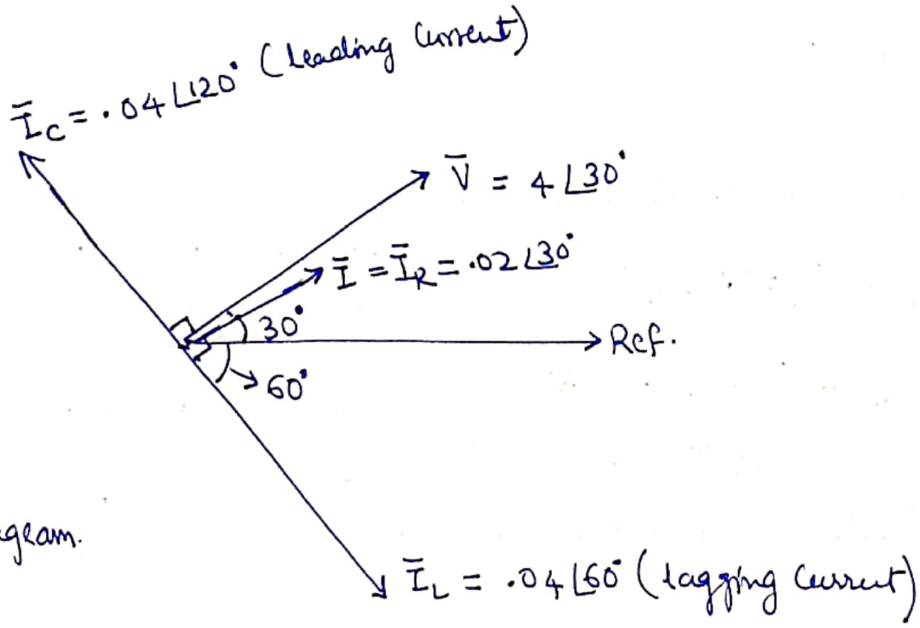
$$\Rightarrow \bar{I}_L = \frac{4\angle 30^\circ}{j200\times 0.5} = .04\angle -60^\circ \text{ A}$$

$$\Rightarrow \bar{I}_C = \frac{4\angle 30^\circ}{\left(\frac{1}{j200\times 50\times 10^{-6}}\right)} = .04\angle 120^\circ \text{ A.}$$

$$\therefore \bar{I} = \bar{I}_R + \bar{I}_L + \bar{I}_C$$
$$= .02\angle 30^\circ + .04\angle -60^\circ + .04\angle 120^\circ$$
$$= (.0173 + j0.01) + (.02 - j0.0346) + (-0.02 + j0.0346)$$
$$= .0173 + j.01 = \underline{\underline{0.02\angle 30^\circ \text{ A}}}.$$

Note that $\bar{I}_L$ & $\bar{I}_C$ phasors, cancell-out in a way so that,
$\bar{I} = \bar{I}_R$ eventually,

The phasor diagram of the same is as follows,



Phasor Diagram.

(b) Since, parallel ckt,

$$\frac{1}{Z_{net}} = \frac{1}{200} + \frac{1}{j200\times 0.5} + \frac{1}{\frac{1}{(j200\times 50\times 10^{-6})}}$$

$$= .005 - j0.01 + j0.01 = 0.005$$

$$\Rightarrow \quad Z_{net} = 200\,\Omega$$

$$\therefore \quad \bar{V} = 200\times 2\angle -40^\circ \quad (\text{Given } \bar{I} = 2\angle -40^\circ)$$

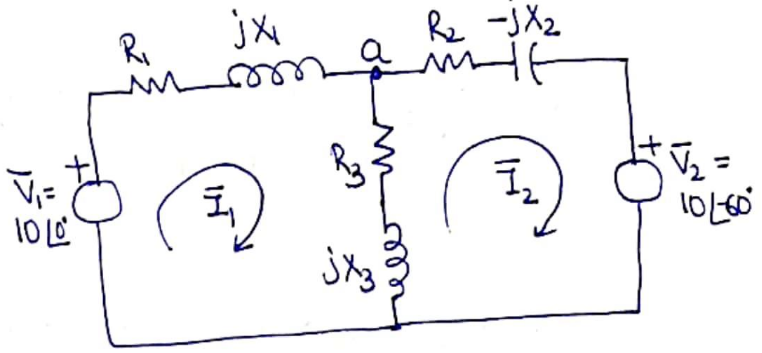
$$= 400\angle -40^\circ$$

$$\Rightarrow \quad \bar{I}_R = \frac{\bar{V}}{R} = \frac{400\angle -40^\circ}{200} = \underline{\underline{2\angle -40^\circ}}$$

& Applied Voltage $\bar{V} = \underline{\underline{400\angle -40^\circ}}$

Network Analysis in 1-ϕ AC Ckts:

Q8. (Mesh Method of Analysis) The two-mesh network with sinusoidal voltage source in each mesh is as shown in Fig. below. The Impedances are given as,



$$\bar{Z}_1 = R_1 + jX_1 = 1 + j = \sqrt{2}\angle 45^\circ$$

$$\bar{Z}_2 = R_2 - jX_2 = 1 - j = \sqrt{2}\angle -45^\circ$$

$$\bar{Z}_3 = R_3 + jX_3 = 1 + j2 = \sqrt{5}\angle 63.5^\circ$$

$$\bar{V}_1 = 10\angle 0^\circ \quad \& \quad \bar{V}_2 = 10\angle -60^\circ$$

Find the phasor expressions for currents through branches $\bar{Z}_1, \bar{Z}_2$ & $\bar{Z}_3$.

A. First step: Write the Mesh Eq^{n}s in terms of Phasor Quantities.

So,
$$\bar{V}_1 - \bar{I}_1\bar{Z}_1 - \bar{Z}_3(\bar{I}_1 - \bar{I}_2) = 0 \quad \rightarrow \text{in Mesh 1}$$

$$\Rightarrow \bar{I}_1\underbrace{(Z_1 + Z_3)}_{Z_{11}} - \bar{I}_2\underbrace{Z_3}_{Z_{12}} = \bar{V}_1 \quad \text{— (1)}$$

Similarly,
$$-Z_3(\bar{I}_2 - \bar{I}_1) - Z_2\bar{I}_2 - \bar{V}_2 = 0 \quad \rightarrow \text{in Mesh 2}$$

$$\Rightarrow -\bar{I}_1\underbrace{Z_3}_{Z_{21}} + \bar{I}_2\underbrace{(Z_2 + Z_3)}_{Z_{22}} = -\bar{V}_2 \quad \text{— (2)}$$

From (1) & (2), upon solving,

$$\bar{I}_1 = \frac{\bar{V}_1 Z_{22} - \bar{V}_2 Z_{12}}{Z_{11}Z_{22} - Z_{12}^2} = \frac{\bar{V}_1(Z_2 + Z_3) - \bar{V}_2(Z_3)}{(Z_1 + Z_3)(Z_2 + Z_3) - Z_3^2}$$

$$= \frac{10\angle 0^\circ (1-j+1+2j) - 10\angle -60^\circ (1+2j)}{(1+j+1+2j)(1-j+1+2j) - (1+2j)^2}$$

Substituting $10\angle 0^\circ = 10$ & $10\angle -60^\circ = (5 - 5\sqrt{3}j)$ & Solving,

we get, $\bar{I}_1 = \frac{20 + 10j - 22.32 - 1.34j}{4+4j} = \frac{8.97\angle 105^\circ}{5.66\angle 45^\circ}$

$$= \underline{\underline{1.58}} \angle 60^\circ$$

$\therefore$ $\boxed{\bar{I}_1 = 1.58\angle 60^\circ \text{ Amp} = 0.79 + j1.37 \text{ A}}$ → Current through Z_1

Similarly, $\bar{I}_2 = \frac{-\bar{V}_2 Z_{11} + Z_{21}\bar{V}_1}{Z_{11}Z_{22} - Z_{12}^2} = \frac{-\bar{V}_2(Z_1+Z_3) + \bar{V}_1 Z_3}{(Z_1+Z_3)(Z_2+Z_3) - Z_3^2}$

Substituting Z_1, Z_2, Z_3 values & $\bar{V}_1 = 10; \bar{V}_2 = 5 - 5\sqrt{3}j$,

we get, $\bar{I}_2 = \frac{34.1\angle 139.3^\circ}{5.66\angle 45^\circ} = 6.03\angle 94.3^\circ$

$\therefore$ $\boxed{\bar{I}_2 = 6.03\angle 94.3^\circ \text{ A} = -0.46 + 6.01j \text{ A}}$ → Current phasor through Z_2

Current phasor through Z_3 is,

$$\bar{I}_1 - \bar{I}_2$$

$$= (.79 + j1.37) - (-.46 + 6.01j)$$

$$= 1.25 - 4.64j = \underline{\underline{4.82\angle -75^\circ}} \text{ Amp.}$$

Q9. (Nodal Method of Analysis). Find the voltage at Node 'a' in the Circuit of 'Q8' with all the same given data.

Also find current in branch Z_3 with this Nodal Analysis.

A. Assuming currents flow away from Node 'a' & applying the Nodal (KCL) Eqn at Node 'a',

$$\frac{\bar{V}_a - \bar{V}_1}{\bar{Z}_1} + \frac{\bar{V}_a}{\bar{Z}_3} + \frac{\bar{V}_a - \bar{V}_2}{\bar{Z}_2} = 0$$

$$\Rightarrow \bar{V}_a \underbrace{\left(\frac{1}{Z_1} + \frac{1}{Z_3} + \frac{1}{Z_2} \right)}_{} = \frac{\bar{V}_1}{Z_1} + \frac{\bar{V}_2}{Z_2}$$

$$\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} = \frac{1}{1+j} + \frac{1}{1-j} + \frac{1}{1+2j}$$

$$= \frac{1}{2} - j\frac{1}{2} + \frac{1}{2} + j\frac{1}{2} + \frac{1}{5} - j\frac{2}{5} = 1.2 - 0.4j$$

$$= 1.263 \angle{-18.4^\circ}$$

&

$$\frac{\bar{V}_1}{Z_1} + \frac{\bar{V}_2}{Z_2} = \frac{10\angle 0^\circ}{\sqrt{2}\angle 45^\circ} + \frac{10\angle{-60^\circ}}{\sqrt{2}\angle{-45^\circ}} = 5 - j5 + 6.81 - j1.83$$

$$\doteq 11.81 - j6.83$$

$$= 13.65 \angle{-30^\circ}$$

Thus,

$$\bar{V}_a \cdot (1.263 \angle{-18.4^\circ}) = 13.65 \angle{-30^\circ}$$

or,

$$\bar{V}_a = \frac{13.65 \angle{-30^\circ}}{1.263 \angle{-18.4^\circ}} = 10.8 \angle{-11.6^\circ} \text{ V}$$

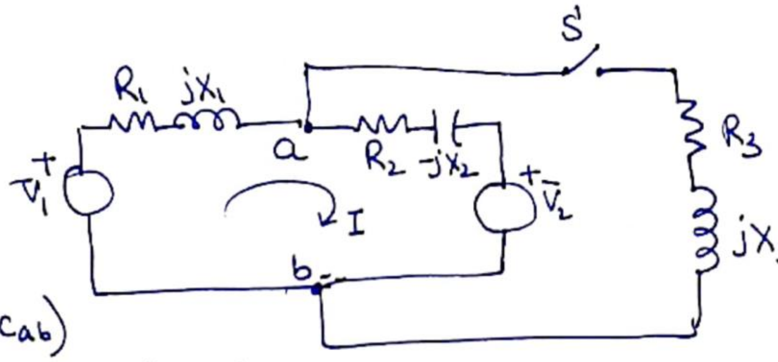
Thus,

$$I_{\text{through } Z_3} = \frac{\bar{V}_a}{Z_3} = \frac{10.8 \angle{-11.6}}{\sqrt{5} \angle 63.5} = 4.82 \angle{-75^\circ}$$

This validates the result in 'Q8' as well.

Q10. (Soln by Thevinin's Theorem) Find the current through the branch 'Z_3' in the Fig. with 'Q8', with all the same given data, by using thevinin's theorem.

A. The circuit can be re-drawn as follows.



As per the Thevinin's theorem, Open Ckt Voltage ($V_{oc_{ab}}$) between 'a' & 'b' must be found. (In order to compute current through Z_3)

So, after opening S', & applying KVL,

$$\Rightarrow \bar{V}_1 - Z_1\bar{I} - Z_2\bar{I} - \bar{V}_2 = 0$$

$$\Rightarrow \bar{I} = (\bar{V}_1 - \bar{V}_2)/(Z_1 + Z_2) = \frac{10 - 10\angle -60^\circ}{1+j+1-j} = \frac{10-5+j8.66}{2}$$

$$= \underline{\underline{5\angle 60^\circ}}$$

$$\therefore V_{oc_{ab}} = \bar{V}_1 - \bar{I} Z_1$$

$$= 10\angle 0^\circ - 5\angle 60^\circ \cdot (\sqrt{2}\angle 45^\circ) = 10 + 1.83 - j6.82$$

$$= 13.65\angle -30^\circ$$

Note: This voltage has larger magnitude than any of source voltages.

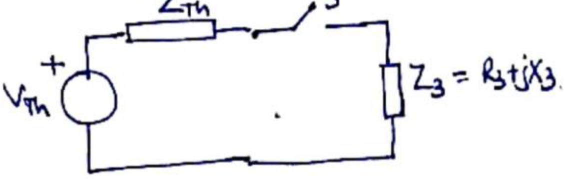
& Thevinin's Impedance: (Z_{Th})

after Short-Ckting voltage sources, the Equivalent Impedance between 'a' & 'b' Can be expressed as,

$$Z_{Th} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{\sqrt{2}\angle 45 \cdot \sqrt{2}\angle -45}{1+j+1-j} = \underline{\underline{1\angle 0^\circ}}\ \Omega$$

$$\therefore \bar{I}_{through\ Z_3} = \frac{\bar{V}_{Th}}{Z_{Th} + Z_3} = \frac{13.65\angle -30^\circ}{1+1+2j}$$

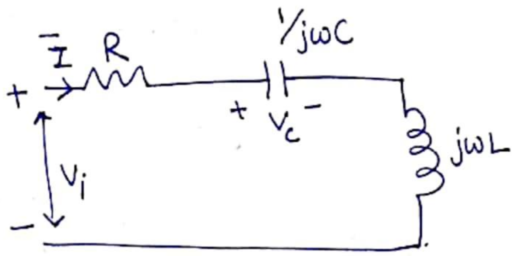
$$= \underline{\underline{4.82\angle -75^\circ}}\ A$$



This validates result in 'Q8' & 'Q9'!

5) Series Resonance:

Q11. A Capacitor of 12 nF is Connected in series with an inductor of 4 mH and 5 Ω resistance as shown below/aside:



ⓐ Calculate Resonant Frequency, ω_0.

ⓑ at ω_0, If $|V_c| = 1.5V$, what is $|\bar{V}_i|$?

ⓒ Along with ~~Using~~ Phasor diagram, show how Capacitor voltage $(\bar{V}_c)$ can be larger than the Input voltage $(\bar{V}_i)$.

A. ⓐ Resonant ~~R~~ Frequency,

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4\times10^{-3}\times12\times10^{-9}}} = 0.144\times10^{6}$$

$$= \underline{144\ \text{k rad/s}}.$$

ⓑ At ω_0, $V_c = 1.5V$

So, $|\bar{I}| = |\bar{V}_c||\omega_0 C| = 1.5\times144\times10^{3}\times12\times10^{-9}$

$= 2.592\,\text{mA}$

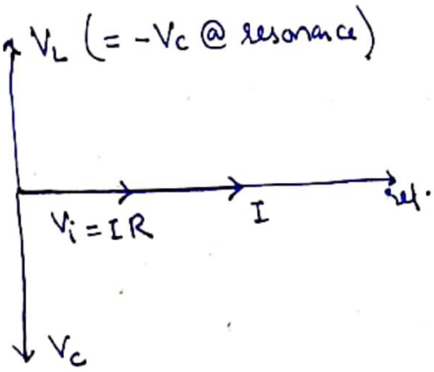
So, $V_i = R\bar{I}$ ($\because$ @ resonance, Circuit Impedance = circuit Resistance)

$= 5\times2.592 = \underline{12.96\,\text{mV}}$.

ⓒ At ω_0, $\bar{I}$ will be in phase with $\bar{V}_i$

& $\bar{V}_c$ lags $\bar{I}$ by 90° ; $\bar{V}_L$ leads $\bar{I}$ by 90°

So,



Capacitive reactance, X_C (or B_C) $= \frac{1}{\omega_0 C} = \frac{1}{144\times10^3 \times 12\times10^{-9}}$

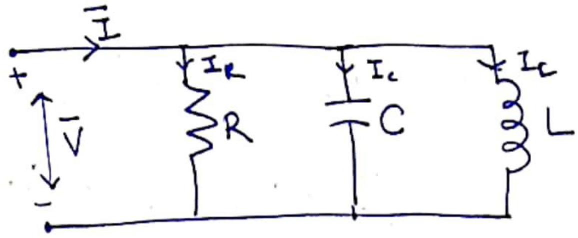
$= 579\,\Omega$ @ resonant Freq. ω_0.

$\therefore$ $X_C >> 5\,\Omega$; Therefore $V_C >> V_i$

6) Parallel Resonance :

Q12. In the circuit as shown in Fig. Aside, $R = \frac{1}{G} = 0.2\,M\Omega$, $L = 2mH$, it draws minimum current at freq. = 5 k rad/s., with min. current = 2 mA.

(a) At what value of 'C', the voltage $\bar{V}$ is maximum? & what is its value?



(b) with the value of 'C' in (a), what should be the value of $\bar{I}$ for inductor current of 1 A

A. (a) For $\bar{V}$ to be maximum, the ckt must be under resonance,

$\Rightarrow \omega_0^2 = \frac{1}{LC}$ or $C = \frac{1}{\omega_0^2 L}$

$\therefore$ $C = \frac{1}{(5\times10^3)^2 \cdot 2\times10^{-3}} = 20\mu F$

Under resonance, $\bar{I}$ flows through 'R' only; So,

$$|\bar{V}| = |\bar{I}|R = |\bar{I}| \times 0.2 \times 10^6$$
$$= 2 \times 10^{-3} \times 0.2 \times 10^6$$
$$= \underline{\underline{400\,V}}$$

(b) Given $L = 2\,mH$

$$\Rightarrow \underset{(=X_L)}{\omega_0 L} = 5 \times 10^3 \times 2 \times 10^{-3} = 10\,\Omega.$$

$$\therefore \bar{I}_L = \frac{\bar{V}}{X_L} = \frac{\bar{V}}{\omega_0 L} = \frac{\bar{I} R}{\omega_0 L}.$$
$$= \frac{0.2 \times 10^6\, \bar{I}}{10}$$

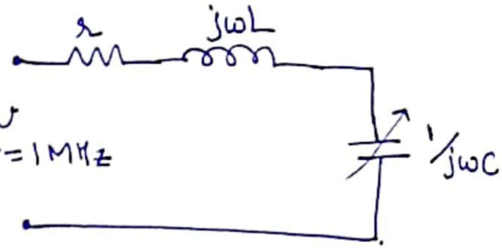
$\therefore$ If $\bar{I}_L = 1A.$, $\Rightarrow \bar{I} = \frac{10}{.2 \times 10^6} \times 1\,A$

$$= 50 \times 10^{-6}\,A = \underline{\underline{0.05\,mA}}.$$

7) Q-Factor (along with Series/Parallel Resonance):

Q13. A voltage (Cnst. Sinusoidal source) with Frequency 1 MHz is connected across an inductor (L) having some internal resistance (r), in series with a variable capacitor. The circuit draws maximum current when the capacitance is 500 pF. The current reduces by a factor of $(1/\sqrt{2})$ when capacitor is adjusted to 450 pF. determine r, L, & Q_0 (Q-Factor).

A. The circuit is drawn as shown aside:



When

$C = 500\,pF$,

Current drawn is max. @ 1 MHz.

i.e. resonant freq. = 1MHz.

$$\Rightarrow \quad \omega_0^2 = \frac{1}{LC} \Rightarrow L = \frac{1}{\omega_0^2 C} = \frac{1}{(2\times\pi\times10^6)^2 \cdot 500\times10^{-12}}$$

$$= .0633\ mH$$

& when $C = 450\ pF$,

the resonant Freq. of ckt becomes, $\omega_0' = \frac{1}{\sqrt{LC}}$

$$= \frac{1}{\sqrt{.0633\times10^{-3}\times450\times10^{-12}}}$$

$$= 5.925\times10^6\ rad/s.$$

$$\Rightarrow f_0' = \frac{\omega_0'}{2\pi} = .943\ MHz$$

∵ With applied frequency of 1MHz, & C=450pF

current reduces by $\frac{1}{\sqrt{2}}$ ∴ ~~f_0' becomes the half power frequency.~~

$\Downarrow$

$$\frac{|I|}{\sqrt{2}} = \frac{|V|}{\sqrt{r^2+\left(\omega L-\frac{1}{\omega C}\right)^2}} \quad ; \quad |I| = \frac{|V|}{r}.$$

@ $\omega =$ ~~2π~~ 1 MHz. and $C = 450\ pF$.

$$\Rightarrow \quad r^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2\frac{|V|^2}{|I|^2} = 2r^2$$

$$\Rightarrow \quad \omega L - \frac{1}{\omega C} = r \quad @ \ \omega = 2\times\pi\times10^6\ rad/s$$

& $C = 450\times10^{-12}\ F$.

while $L = .0633\ mH$

$$\Rightarrow \quad 2\pi \times 10^6 \times .0633 \times 10^{-3} - \frac{1}{2\pi \times 10^6 \times 450 \times 10^{-12}} = r$$

$$\Rightarrow \quad r \approx 44\,\Omega$$

Quality Factor, $Q_0 = \dfrac{\omega_0 L}{R}$

$$= \frac{2\pi \times 10^6 \times .0633 \times 10^{-3}}{44} \approx 9.$$

Q14. An RLC parallel circuit has $R = 1\,M\Omega$, $L = 1H$, $C = 1\mu F$. It is excited by a current $I = 10\angle 0^\circ \mu A$. Find $\underset{\text{Resonant Freq.}}{\underset{\downarrow}{\omega_0}}$ and Q_0.

If freq. of current $\bar{I}$ is ω_0, find voltage accross the circuit. At what frequencies, the voltage reduces by $(1/\sqrt{2})$.?

A.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 1 \times 10^{-6}}} = 10^3 \text{ rad/s.}$$

At resonance, current $\bar{I}$ flows only through 'R'. Therefore,

$$\bar{V} = \bar{I} R = 10 \times 10^{-6} \times 1 \times 10^6 = 10\angle 0^\circ \text{ V}$$

$\uparrow$

voltage accross ckt when $\bar{I}$ has freq. ω_0.

→ The voltage gets reduced by $(1/\sqrt{2})$ at the Half-power frequencies

Therefore,

Firstly, Quality factor $Q_0 = \omega_0 RC$ (In parallel RLC circuit)

$$= 10^3 \times 1 \times 10^6 \times 1 \times 10^{-6}$$

$$= \underline{\underline{10^3}}.$$

Band-width,

$$\omega_b = \frac{\omega_0}{Q_0} = \frac{10^3}{10^3} = 1 \text{ rad/s}.$$

$\therefore$ Frequencies at which voltage reduces by $1/\sqrt{2}$ is,

$$\omega_0 \pm \frac{\omega_b}{2}$$

$$\therefore \quad \omega_1 = \omega_0 - \frac{\omega_b}{2} = 10^3 - \frac{1}{2} = \underline{999.5} \text{ rad/s.}$$

$$\omega_2 = \omega_0 + \frac{\omega_b}{2} = 10^3 + \frac{1}{2} = \underline{\underline{1000.5}} \text{ rad/s}$$

———————————— x ————————————

Best Wishes

— ☺ Wedhimoni